CS & IT ENGINEERING

2025

Theory of Computation

{GATE} Marks (7-10 ✓) 10 ✓

Logical

Scoring

{{TOC + Compiler}} (15-17)

DFA

Lecture No.- 01

By- Venkat sir

Recap of Previous Lecture

1

Topic Introduction of TOC

Topic faculty

Name:

1. Lecture
2. DPP ✓
3. Test ✓
4. GATE PYQ ✓

{80%}

THEORY OF COMPUTATION

{GATE}

10

5 Topic ① 50% — Finite Automaton & Regular Languages. Regular Expression

3 Topic ② 30% — Pushdown Automata & Context free Languages. & Context free Grammar

Topic ③ — Turing Machine & Recursive Enumerable Languages.

20%

and

Topic ④ — Undecidability.

{D.M} {Set}

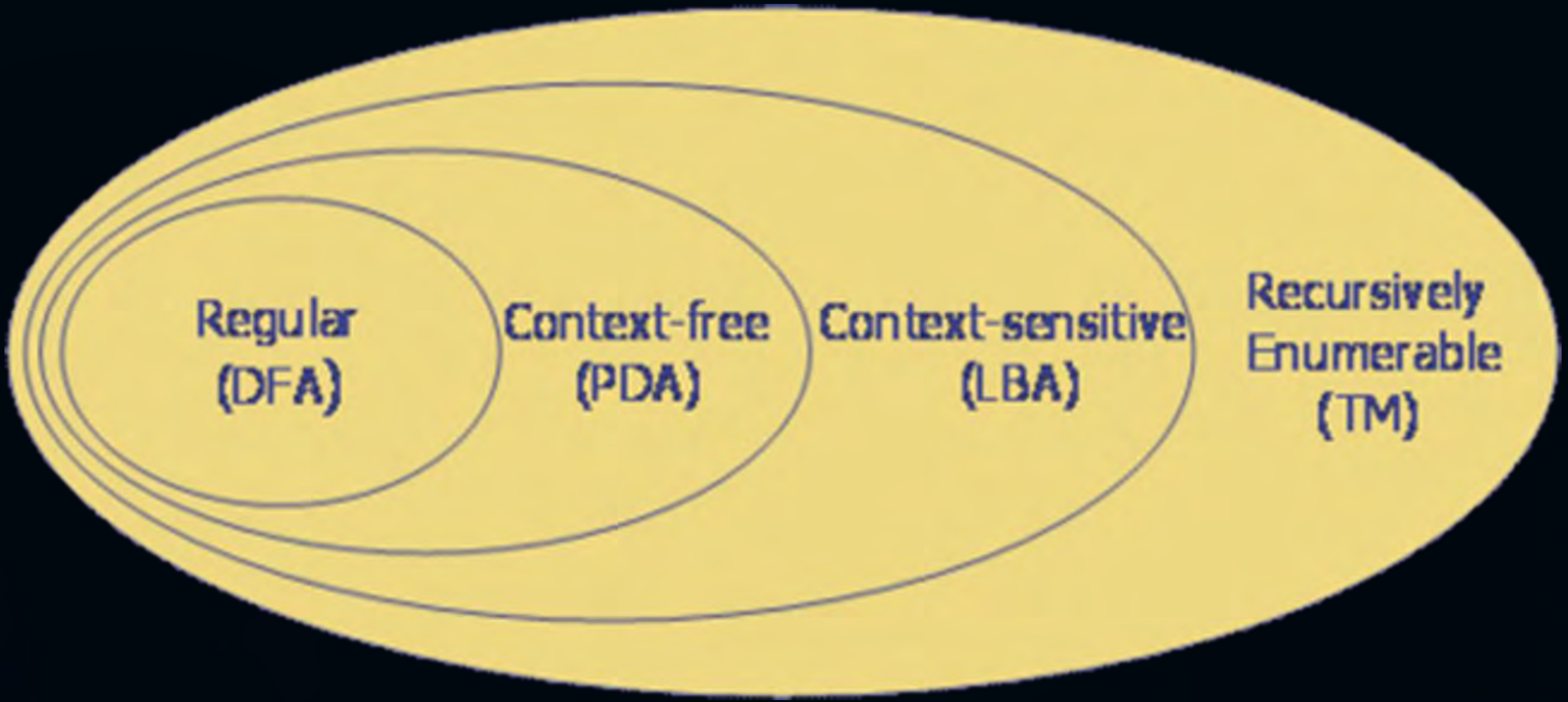
Topic : Theory of Computation

{D.M}

Reference Books

① Ullman

② Peter Linz



Topic : Introduction:

{GATE}

{FLAT}
{TCS}
{A&L}

~~It~~ TOC is the mathematical study of computing machines and their capability

(or)

It is the study of automata theory and formal languages.

machine

C, c++, java . .

1. Finite Automata ⟷ Regular Language
2. Pushdown Automata ⟷ Context free Language
3. Linear bounded Automata ⟷ Context sensitive language
4. Turing machine ⟷ Recursive Enumerable Language

Topic : Introduction:

Computer can solve it

Decidable Problem : { Algorithm exist }

Profes

Undecidable Problem : { No Algorithm exist }

more

Computers can't solve this

Topic : Terminologies:

Alphabet(Σ): Finite non-empty set of symbols

Ex:- {a, b} –

{a, 1, 2} –

{ } –

Topic : Terminologies:

Σ

set = {1, 2, 3}

Alphabet(Σ): Finite non-empty set of symbols

any

Ex:-

1. Σ = {a, b} – ✓
2. Σ = {a, 1, 2} – ✓
3. Σ = { } – ✗
4. Σ = {अ, आ} ✓
5. Σ = {6, 0, 2}

Topic : String:

String: Finite sequence of symbols over the given alphabet Σ.

"Hi"

" "

$\Sigma = \{a, b\}$

a

$|ab| \rightarrow 2$

$aaa \rightarrow 3$

$|abab| \rightarrow 4$

$a \rightarrow 1$

$abaabbaa \rightarrow$

Ex

$a^0 \rightarrow \epsilon$

$b^0 \rightarrow \epsilon$

Epsilon

zero length string

$a^1 \rightarrow a$

$a^2 \rightarrow |aa|$

$a^3 \rightarrow aaa$

$a^0 b^0 \rightarrow \epsilon$

$a^2 b^2 \rightarrow aabb$

Topic : String:

$\Sigma = \{a, b\}$ formal language empty set

Language :- Any set of strings over the given alphabet $\Sigma = \{a, b\}$.

$L_1 = \{ ab , ba , abab \}$ - finite language

$L_2 = \{ a , ab , aba , ... \}$ - Infinite Language

$L_3 = \{ \}$ - valid $\Rightarrow$ empty language

$L_4 = \{\epsilon\}$ – finite Language

$L_5 = \{ a \}$ $\rightarrow$ finite language

$L_6 = \{ \epsilon , a , b , aa , ab , ba , bb ------ \}$ - infinite language

$\boxed{\text{Complete Language}}$

$L_7 = \{\epsilon, a, b\} \Rightarrow$ finite Lang

$L_3 = \{ \}$ Empty

$L_4 = \{\epsilon\}$ finite

Topic : String:

Sub – String : Consecutive sequence of symbols over the given string.

Total no of substring for the given string $= n(n + 1)/2 + 1$

1

FINITE AUTOMATA

and

Regular Language

Topic : Finite Automata

$L = \{\epsilon, aa, ba, \ldots\}$

It is a mathematical model which contains finite number of states and transitions.

Finite state machine

Finite Automata

q_0

input → **Without Output**

1. DFA ***
2. NFA
3. **Є - NFA**

input → **With output** → output

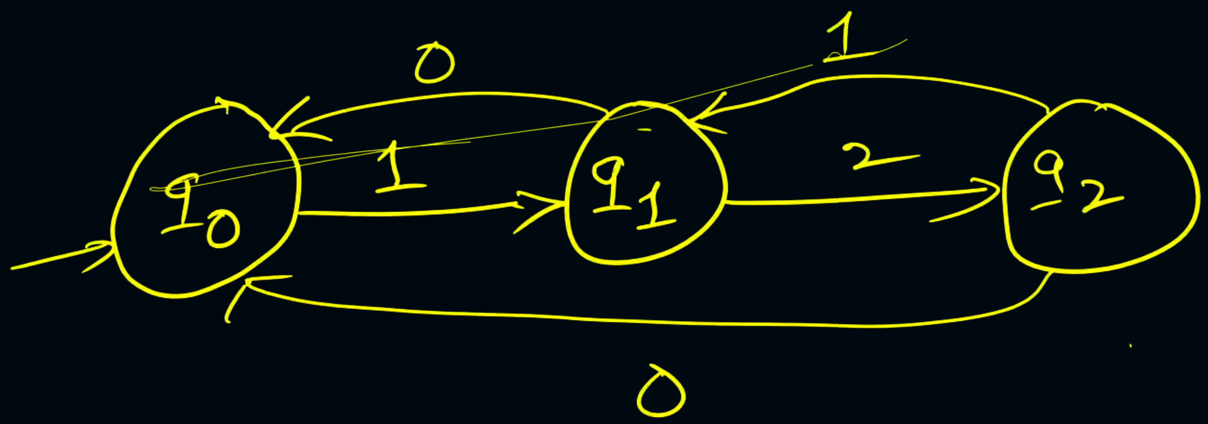
4. Mealy machine
5. Moore machine

Circuit machine

q_0 011 110

FAN

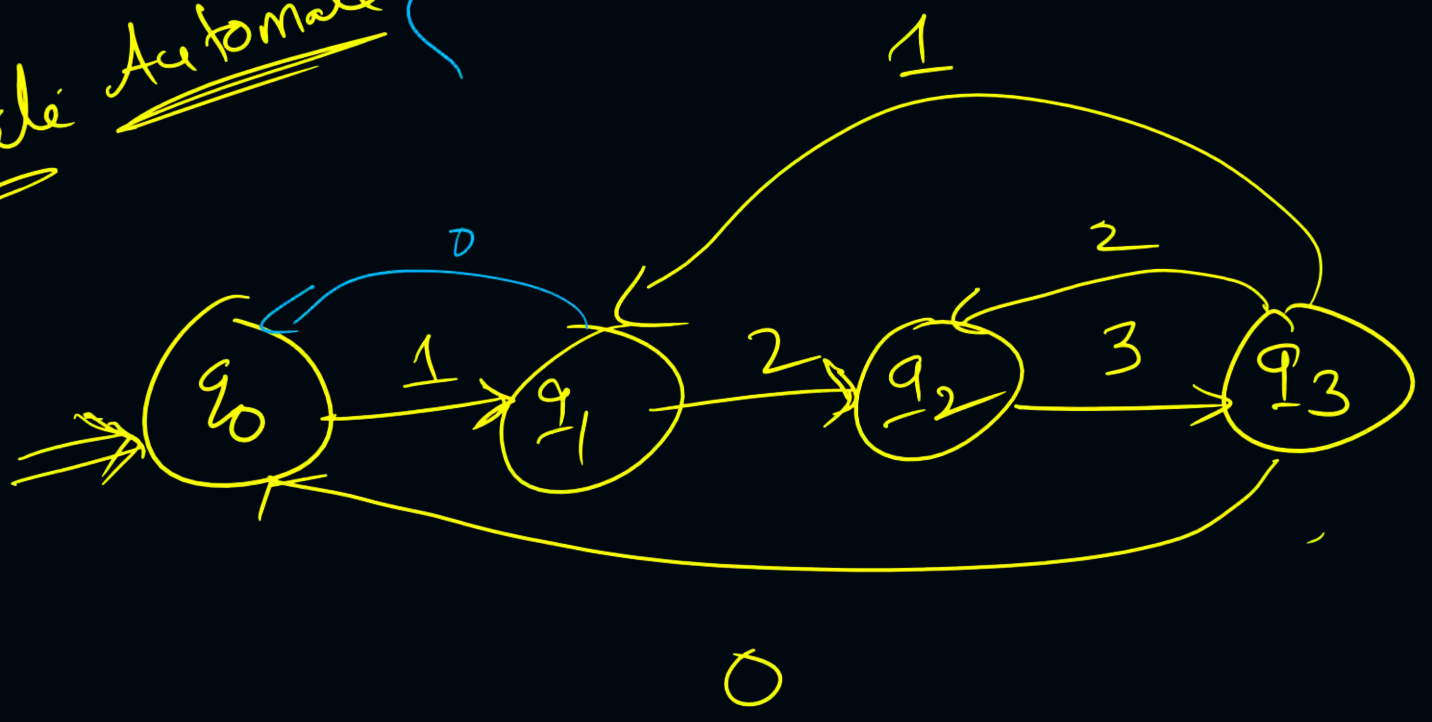
Finite Automata



3 States

Bike

{Finite Automate}



Topic : Deterministic Finite Automata

DFA : It is a finite automata in which from every state on every input symbol exactly one transition should exits.

FORMAL DFA :

DFA is defined as

$$DFA = (Q , \Sigma , q_0 , F , \delta)$$

Q : Finite set of states

Σ : Input alphabet

q_0 : Initial state

F : Set of final states

δ : Transition function $Q * \Sigma \rightarrow Q$

Topic : Deterministic Finite Automata

DFA

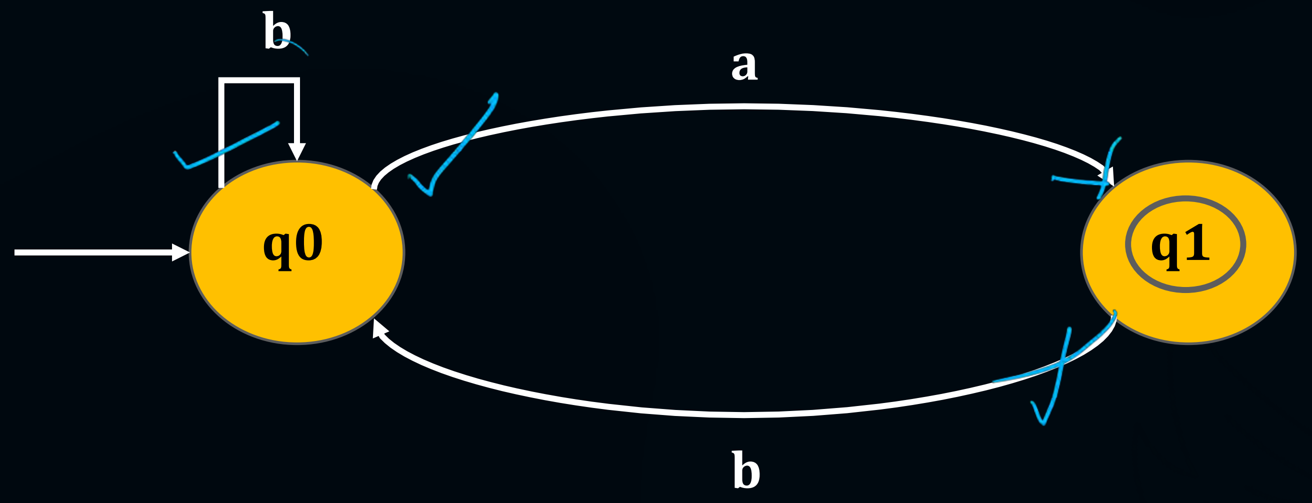
Example of DFA ?

$\Sigma = \{a, b\}$

not a DFA

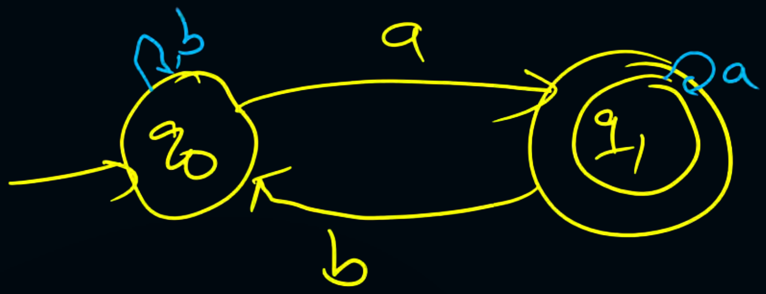
$q_1 \to a \to$ missing

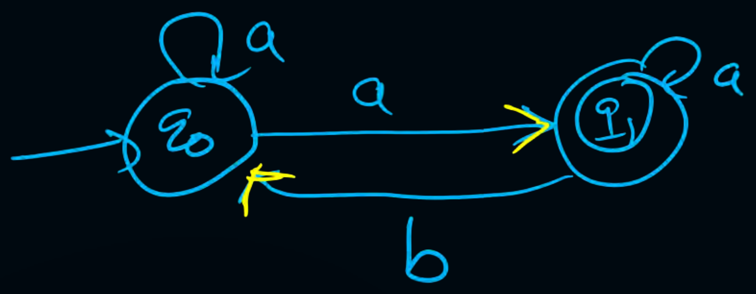
(1)



$\Sigma = \{a, b\}$

not a DFA ?

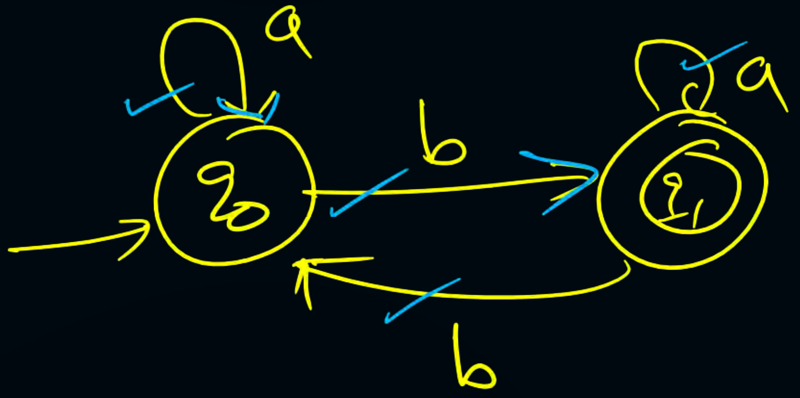




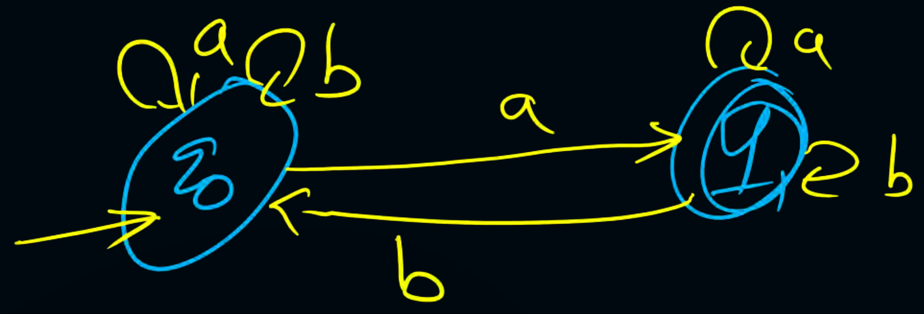
not DFA

$\times \quad \delta(q_0, a) = \{q_0, q_1\}$

$\delta(q_0, b) = \times$



DFA ✓



not a DFA

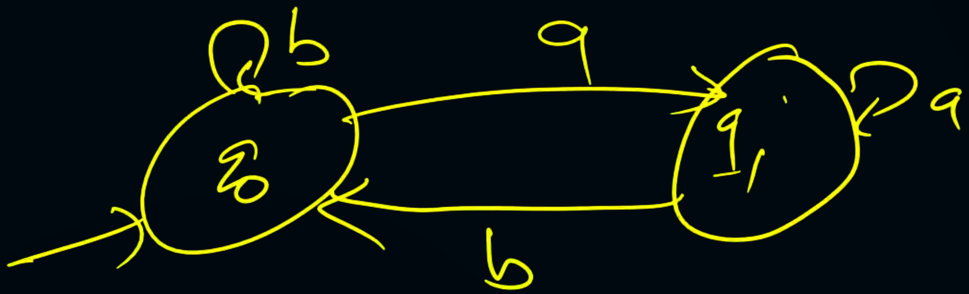
$$(q_0, a) = \{q_0, q_1\} \times$$

$$\delta(q_1, b) = \{q_0, q_1\} \times$$

Home work

$\Sigma=\{a,b\}$

1



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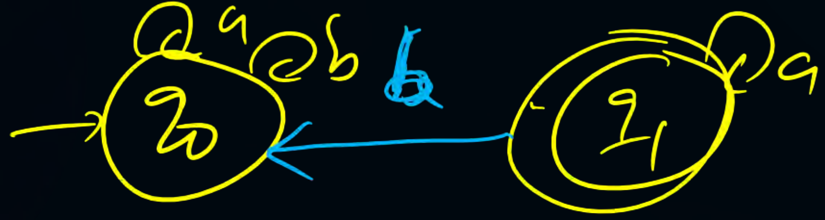
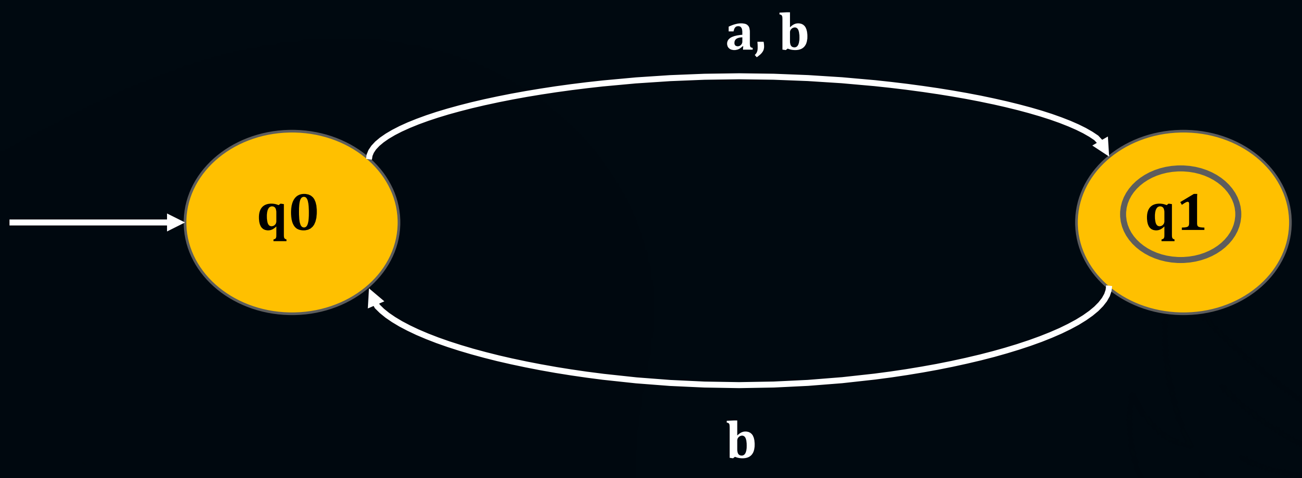


Topic : Deterministic Finite Automata

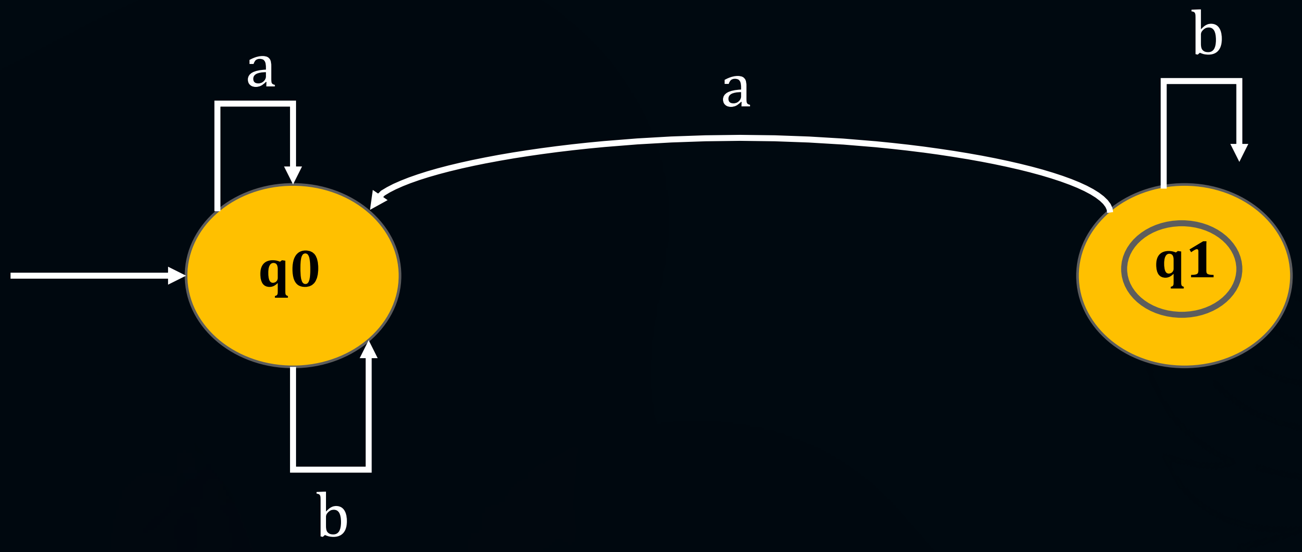
Example of DFA :

(2)



Example of DFA :

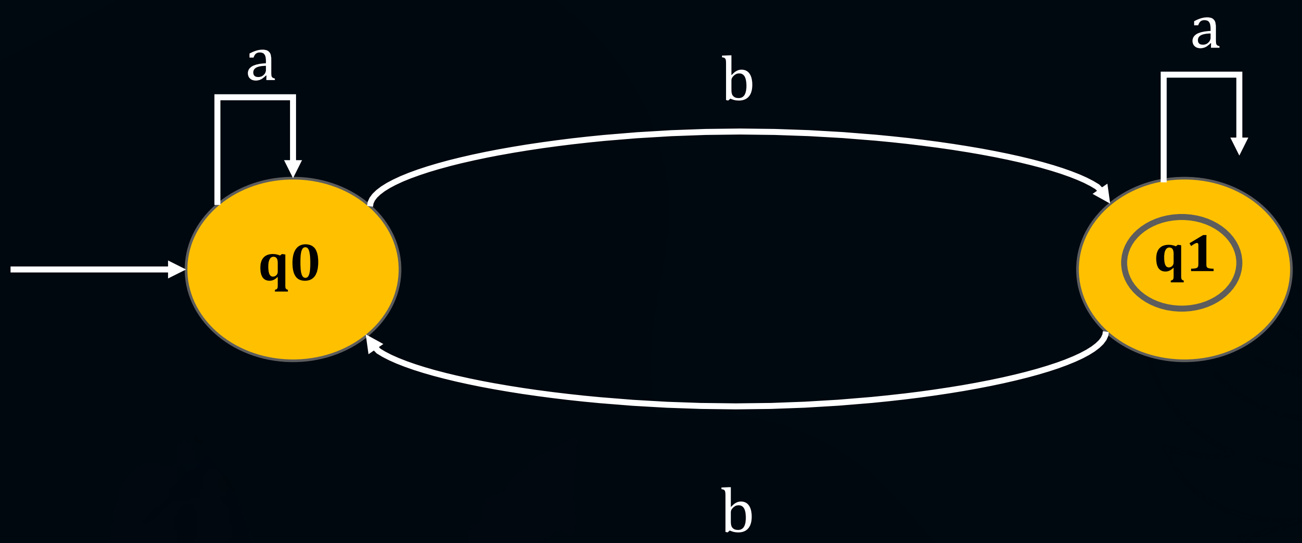
(3)



Topic : Deterministic Finite Automata

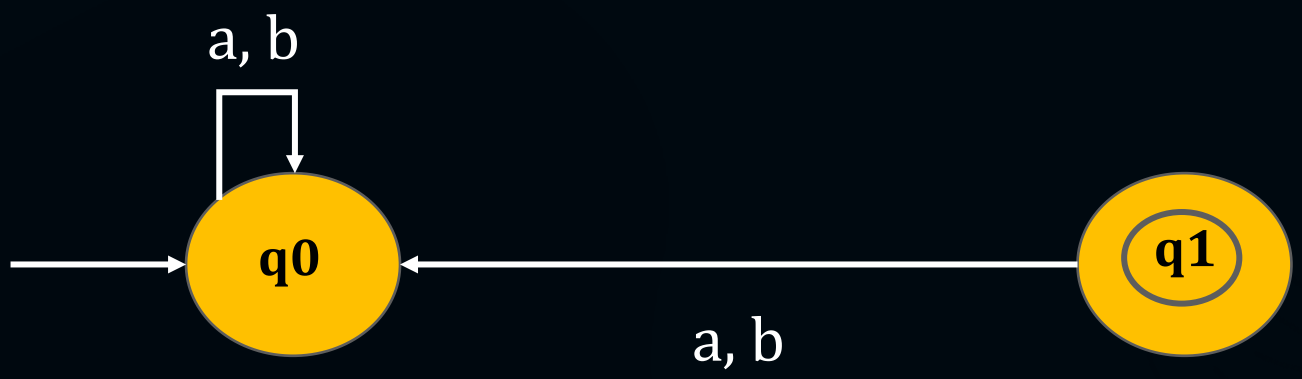
Example of DFA :

(4)



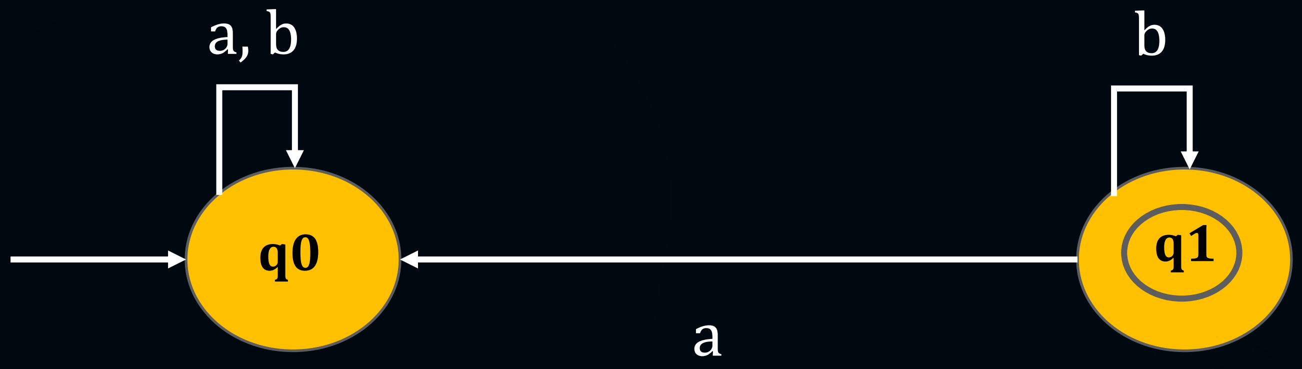
Example of DFA :

(5)



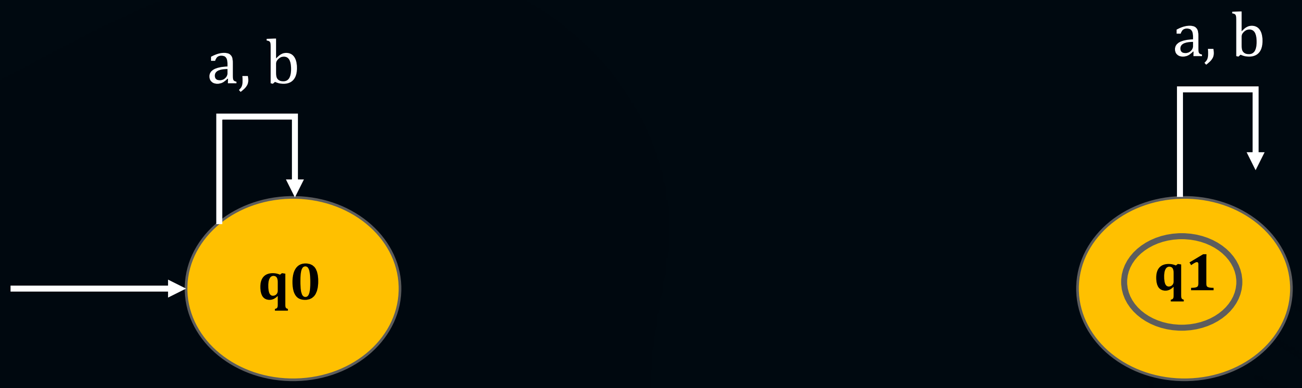
Example of DFA :

(6)



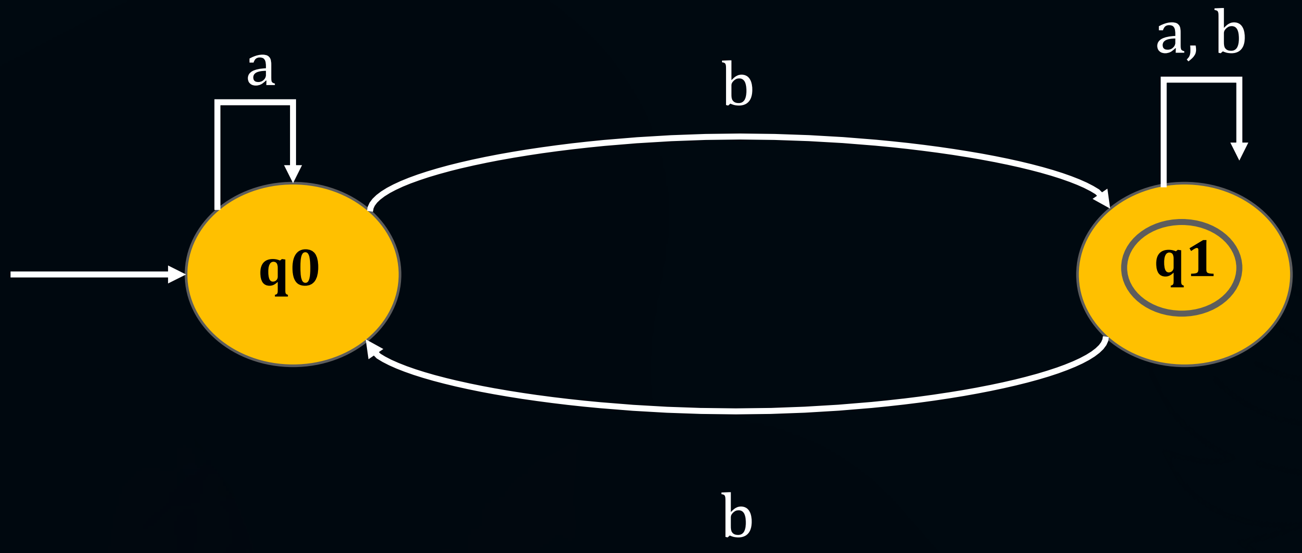
Example of DFA :

(7)



Example of DFA :

(8)



THANK - YOU